

Tarsha de Souza

Senior Software Engineer

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Seven years a software engineer, primarily iOS, branching out wherever there's been room — a six-month rotation on Expedia's marketing ML team, open source in my own time. I like to build and fix things, I like work with room to be creative, and most of all I like learning something I didn't know.

EXPERIENCE

Expedia Group — Senior Software Engineer — Design Systems Platform

Dec 2020 — Present · Madrid, Spain

The design-system platform 23 brands build on, plus work outside it.

AI engineering — 2024 to present

- Built multi-agent systems (Claude, Codex) for design-to-code generation from Figma, autonomous debugging and self-healing CI — agent-driven fixes gated by verification loops and human-in-the-loop checkpoints.
- Created reusable Claude Code skills and orchestration agents adopted across the org through an internal skills marketplace.

ML engineering — six-month rotation with the marketing ML team, Feb — Aug 2026

- Refactored the production cost/volume circuit breaker in a churn-prevention PySpark pipeline from hardcoded brand logic to a config-driven market registry — new markets now onboard through a YAML change with zero code changes, verified against production data on Databricks and the live Airflow job.
- Found and fixed silent data-quality bugs: UK traffic skipped by region detection, crashes on legacy tables, and a rerun bug blocking every market. Wrote the architecture docs and onboarding runbook for the team.
- Added regression handling for bad days — a skewed dataset or a late backfill now raises an alert carrying the context needed to diagnose it, rather than passing silently.

iOS & design systems

- Cut iOS build times by 24% through modularisation and tooling, after profiling the build graph.
- Built server-driven UI authentication flows on GraphQL, so stakeholders ship login-flow changes across 23 brands without waiting for an app release cycle.
- Built the automated maturity-tier promotion system for the next-generation design-system iOS library, and set up its Artifactory release pipeline.
- Built App Intents backed by live data in a five-day hackathon and took them to production — 130,000+ events between 25 September and 7 October 2024, 120,000 of them from the lodging intent alone. Still shipping in the Expedia app.
- Created Xray, an internal debugging tool for design-system components.

Banco Santander — Software Engineer (via consultancy)

2017 — 2019

Customer-facing features in a large-scale, regulated banking app.

- Integrated a conversational virtual assistant into the banking app.

TEACHING AND SPEAKING

- Building the ultimate traveller companion — Women of Silicon Roundabout, Nov 2022 · London
- GraphQL best practices — Product Academy — Expedia Group, 2023 · taught in India, London, Chicago, Seattle, Austin, San Francisco

EDUCATION

- Biomedical Science, Cumberland University · 2014 — 2017